

ORIGINAL ARTICLE

Impact of Constipation Therapies on Severity of Gastroparesis and Constipation Symptoms in Relation to Gastric and Colonic Transit

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ABSTRACT

Introduction: Studies demonstrate an overlap of constipation with gastroparesis and functional dyspepsia, but the impact of treatments that target constipation on improving upper gastrointestinal (UGI) symptoms is unexplored. We quantified the effects of constipation medication therapies on UGI and constipation symptom severity in subjects presenting with symptoms of gastroparesis.

Methods: Fifty-six subjects with symptoms of gastroparesis underwent concurrent wireless motility capsule and gastric emptying scintigraphy and were recommended to receive either a new medication therapy for constipation or a change in constipation therapy based on investigator interpretation of test results. Gastroparesis Cardinal Symptom Index (GCSI), upper abdominal pain, and constipation scores were compared between baseline and 6 months. Data were compared between delayed or non-delayed gastric emptying and the presence or absence of slow colonic transit.

Key Results: Subjects with slow colonic transit had improvements in GCSI ($p=0.007$) and constipation scores ($p=0.004$) after treatment with a new or changed constipation medication, with the delayed emptying subgroup driving GCSI improvements ($p=0.004$). Reductions in nausea/vomiting ($p=0.02$) and early satiety/fullness subscores ($p=0.002$) with trends to improved bloating/distention subscores ($p=0.06$) were observed in this subgroup, but upper abdominal pain was unchanged. Subjects with normal colonic transit showed no improvement in GCSI scores regardless of gastric emptying status ($p>0.05$).

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Conclusions and Inferences: Identifying and treating delayed colonic transit in gastroparetic (delayed gastric emptying) subjects improves global UGI symptoms as well as selected individual symptoms. Evaluation of whole gut motility as well as recognizing and managing extragastric delay may promote improved outcomes in these patients.

Trial Registration: [ClinicalTrials.gov](https://clinicaltrials.gov): NCT02022826

1 | Introduction

Upper gastrointestinal (UGI) symptoms of nausea, vomiting, early satiety, postprandial fullness, bloating, distention, and epigastric pain can reflect underlying gastroparesis or functional dyspepsia, affecting up to an estimated 7% of the worldwide population. These patients consume a significant amount of health-care resources, stemming from patient visits, diagnostic testing, treatment, and surgery, that have been estimated at \$18.4 billion in 2009 [1, 2]. Approximately 15.8% require more than five visits to gastroenterology clinics, and 53.9% had emergency department presentations to manage symptoms within a 1-year observation period for gastroparesis and nausea/vomiting syndromes [3, 4]. Guideline-recommended treatments for these symptoms typically include dietary modifications, proton pump inhibitors, antiemetics, prokinetics, and visceral pain management with neuromodulators [5, 6]. However, management remains imperfect, and the challenges stem from the heterogeneity of presentation for most gastrointestinal motility disorders. One facet of clinical evaluation and management that has been underappreciated in this patient population is the occurrence of coexisting lower gastrointestinal dysmotility [7].

There is increasing recognition that focusing solely on gastric emptying delays when making treatment decisions does not take into consideration the range of pathophysiologic abnormalities in patients with suspected gastroparesis. UGI symptoms often overlap with lower gastrointestinal symptoms of constipation. Moderate to very severe constipation symptoms, as measured by the Patient Assessment of Upper Gastrointestinal Disorder Symptoms survey, are prevalent in 58% of patients with gastroparesis [7]. Further, an estimated 11.6% of patients with functional dyspepsia complain of constipation [8]. Multiple diagnostic modalities, including whole-gut scintigraphy, radiopaque sitz-markers, and wireless motility capsules (WMC) detect slow colonic transit in up to one-third of patients presenting with gastroparesis symptoms, whether in isolation or in combination with other regional gut delays [7, 9–11]. In appropriate clinical situations, a broader whole-gut management approach may be necessary [12–14]. Specifically, identification of colonic delay has been shown to influence treatment decisions directed towards lower gastrointestinal (LGI) therapies [7, 15, 16].

Therefore, evaluation for slow colonic transit may prove to be a useful tool in the clinical care of patients with symptoms suggestive of gastroparesis. However, the clinical impact of this finding and whether treating lower-gut disorders such as slow colonic transit affects UGI symptoms is currently unknown. We hypothesize that, in patients presenting with primarily gastroparetic or functional dyspeptic symptoms, identifying and treating slow colonic transit improve UGI symptoms. Our aims in this secondary analysis of a prospective cohort study were to assess the clinical impact of identifying and treating slow colonic transit in

patients presenting with UGI symptoms suggestive of gastroparesis and how their treatment outcomes were influenced by their gastric or colonic transit status.

2 | Materials and Methods

2.1 | Subject Enrollment

One hundred sixty-seven subjects 18–80 years old presenting with ≥ 2 UGI symptoms suggestive of gastroparesis (nausea, vomiting, upper abdominal pain, early satiety, bloating, postprandial fullness) for ≥ 12 weeks as their primary complaint were enrolled in a clinical trial conducted from 2013 to 2016 ([ClinicalTrials.gov](https://clinicaltrials.gov): NCT02022826) at 10 centers. Upper endoscopy or radiology excluded organic disorders. Exclusion criteria included dysphagia, previous gastrointestinal surgery (except appendectomy, cholecystectomy, hysterectomy), any abdominopelvic surgery within 3 months, inflammatory bowel disease, chronic nonsteroidal anti-inflammatory drug use, diverticulitis, intestinal strictures, hemoglobin A1c $> 10\%$, implanted cardiac devices, and body mass index (BMI) $> 40 \text{ kg/m}^2$. The protocol was approved by the Institutional Review Board at each center. Informed consent was obtained from each subject before enrollment.

2.2 | Study Protocol

This investigation presents post hoc analyses of prospectively collected data to assess the impact of treatment with constipation therapies on UGI and LGI symptoms from a parent study in which subjects with suspected gastroparesis underwent concurrent WMC and gastric emptying scintigraphy (GES) testing. Subjects confirmed discontinuation of constipation medications, proton pump inhibitors, histamine₂ receptor antagonists, prokinetic agents, opioid agents, anticholinergic drugs, and cannabinoids at least 72 h prior to testing as part of the inclusion criteria.

The WMC system (Medtronic, Minneapolis, MN) included a wireless receiver, display software, and a single-use capsule with sensors for pH, temperature, and pressure that permit regional gut transit and intraluminal pressure evaluation [17]. Subjects ingested a standardized egg sandwich meal (255 kcal—72% carbohydrates, 24% protein, 2% fat, 2% fiber) with 120 mL water following overnight fasting and then swallowed the WMC with an additional 50 mL of water. Eight hours after capsule ingestion, all subjects consumed a second meal consisting of 250 mL Ensure (Abbott Laboratories, Illinois, USA) per protocol. Nine hours after capsule ingestion, subjects were allowed to freely take in liquids and solid food. Subjects left the study center with the data receiver for continuous WMC data acquisition. Five to 7 days after WMC ingestion, subjects came back to the study center to return the data receiver and the diary.

Summary

- Tailored constipation therapy improves Gastroparesis Cardinal Symptom Index (GCSI) scores in subjects with gastroparesis symptoms that have delayed colonic transit, particularly in those with concomitant gastric delay, but not in those with normal colonic transit irrespective of gastric transit status.
- Significant improvements in GCSI sub-scores of nausea/vomiting and early satiety/fullness but not bloating/distention or upper abdominal pain were further observed.
- Whole-gut transit testing in suspected gastroparesis may offer clinical insight to identify patients with colonic dysmotility, thereby expanding therapeutic approaches beyond a gastric focus.

Five to 28 days after testing, site investigators made recommendations for management changes (including constipation medications, prokinetics, and neuromodulators) after review of WMC and GES transit testing results during an outpatient study visit. Management decisions were made at the discretion of each site investigator based on his/her typical clinical practice pattern. This protocol design was purposeful with the intention to mimic real-world practice inclusive of assessing the impact of transit test findings on clinical decision making. A change in constipation medication(s) was defined as a start of a new medication therapy or a change in an existing constipation medication compared to baseline. Subjects not meeting criteria for a change in constipation medication or subjects without any constipation medications were excluded from these analyses. Slow colonic transit was defined as a WMC colon transit time (CTT) > 59 h [17]. Delayed gastric emptying was defined as a WMC gastric emptying time (GET) > 5 h. Taken together, these results were used to categorize subjects as either delayed or normal CTT and delayed or non-delayed GET as the primary factors defining groups of interest for evaluating response to constipation management changes. Data on age, gender, BMI, Bristol Stool Form (BSF) scale (with scores of 1 or 2 representative of constipation), satisfaction of criteria for functional dyspepsia and/or constipation based on validated Rome III modules, prokinetic and neuromodulator medication use, and constipation medication use were collected at baseline [18]. The Rome III constipation criteria and the BSF were used to describe the cohort at baseline but were not followed longitudinally at subsequent study visits.

Gastrointestinal symptom severity at baseline prior to transit testing and 3 and 6 months after transit testing was quantified using the Gastroparesis Cardinal Symptom Index (GCSI) [19]. Symptom scores for upper abdominal pain and constipation were obtained from a modified Patient Assessment of Upper Gastrointestinal Disorders—Symptom Severity Index (PAGI-SYM) also administered at baseline prior to, as well as 3 and 6 months after transit testing [20]. Overall GCSI scores, GCSI subscores for nausea/vomiting, early satiety/fullness, and bloating/distention, as well as upper abdominal pain and constipation scores from the PAGI-SYM were rated on a severity scale from 0 to 5 defined as 0—none, 1—very mild, 2—mild, 3—moderate, 4—severe, and 5—very severe. These measures served as

the outcome variables of interest in evaluating responses to constipation management changes.

2.3 | Statistical Analysis

Age, BMI, overall GCSI, and constipation severity scores at baseline, and WMC GET and CTT values are presented as mean (standard deviation or SD) and were compared using two-sided Student *t*-tests in those with delayed versus non-delayed GET within CTT classifications. Gender, the presence of diabetes, GES delay status, functional dyspepsia, and constipation by Rome III module testing, BSF scores of 1 or 2 (reflecting constipation), and baseline constipation medication use are presented as number (%) and were compared using Fisher's Exact Test.

A generalized linear model using repeated measures analysis of variance (ANOVA) compared symptom scores at baseline and 3 and 6 months. All analyses presented in this article include only the baseline and 6-month values on symptom severity to focus on the longer timeline that is more reflective of durable outcomes. The *p*-values are derived from the post hoc pairwise comparisons performed after the repeated measures ANOVA. Symptom scores assessed included overall GCSI, individual GCSI subscores for nausea/vomiting, early satiety/fullness, and bloating/distention, as well as the upper abdominal pain and constipation symptom scores from the modified PAGI-SYM. Adjustments in the model were made individually for age, sex, BMI, functional dyspepsia, delayed gastric emptying on GES, prokinetic use, and neuromodulator use when the main model showed significant differences in overall GCSI scores and constipation symptom scores. Overall GCSI scores at baseline and 6 months are plotted as median with interquartile range (IQR) and maximum and minimum values for continuous variables on box and whisker plots. Changes in ordinal variables including individual symptom scores of the GCSI and subscores for upper abdominal pain and constipation are plotted as the mean difference $\pm$ SD from 6-month values minus values at baseline instead of median (IQR) values. This permitted greater visual discrimination of the magnitude of response to constipation therapies between CTT and GET subgroups on graphical presentation. To further assess the benefits of constipation medications on constipation symptom severity, subgroup analyses defined percentages of patients with reductions in constipation scores in those with delayed versus non-delayed GET and of those not on constipation medications at the baseline visit and were compared using Fisher's Exact Test. Two-sided *p*-values with a nominal statistical significance of *p* < 0.05 were used. Analysis was performed using IBM SPSS v29.0.0.0 software (Armonk, NY).

3 | Results

3.1 | Subject Characteristics

Management plans were completed by 150 of 167 enrolled subjects. Three of the 150 subjects had severely prolonged GET, precluding assessment of CTT, and 12 subjects did not complete 6-month follow-up surveys, leaving 135 subjects available for analysis of their data. Fifty-six subjects from this remaining sample received either a new constipation medication or a change in constipation therapy, and this specific cohort was the

TABLE 1 | Patient characteristics.

	Delayed CTT			Normal CTT		
	Delayed GET <i>n</i> = 15	Non-delayed GET <i>n</i> = 21	<i>p</i>	Delayed GET <i>n</i> = 4	Non-delayed GET <i>n</i> = 16	<i>p</i>
Age (years)	46.3 ± 13.0	48.8 ± 18.3	0.66	34.8 ± 13.5	50.0 ± 14.4	0.07
Gender			0.37			0.51
Male	1 (6.7%)	5 (23.8%)	—	1 (25.0%)	2 (12.5%)	—
Female	14 (93.3%)	16 (76.2%)	—	3 (75.0%)	14 (87.5%)	—
BMI (kg/m ²)	24.5 ± 4.07	29.4 ± 5.75	0.008	25.6 ± 10.3	28.5 ± 7.21	0.51
Diabetes	3 (20.0%)	8 (38.1%)	0.30	0 (0.0%)	4 (25.0%)	0.54
Functional dyspepsia by Rome III module	11 (78.6%)	15 (71.4%)	0.71	3 (100%)	13 (81.3%)	1.00
GES delay	7 (46.7%)	2 (9.5%)	0.02	1 (25%)	1 (6.3%)	0.37
WMC GET (hours)	13.6 ± 16.2	3.54 ± 0.95	0.007	12.6 ± 7.71	3.02 ± 1.01	< 0.001
Overall GCSI score, baseline	3.21 ± 0.55	2.67 ± 0.92	0.05	2.67 ± 0.84	2.55 ± 1.24	0.86
Constipation by Rome III module	8 (53.3%)	9 (42.9%)	0.74	1 (33.3%)	5 (31.3%)	1.00
WMC CTT (hours)	89.5 ± 22.5	96.4 ± 26.8	0.42	20.2 ± 18.8	32.4 ± 13.4	0.15
BSF score 1 or 2, baseline		18 (50.0%)			5 (25.0%)	
Constipation score, baseline	3.33 ± 1.59	3.71 ± 1.27	0.43	2.75 ± 0.50	3.38 ± 1.75	0.49
On constipation medication, baseline	6 (40.0%)	11 (52.4%)	0.52	1 (25%)	6 (37.5%)	1.00

Note: Continuous variables represented as mean (standard deviation) and categorical variables represented as number of subjects (percentage of total). Functional dyspepsia and constipation as defined by Rome III criteria. Bold values are merely to highlight the significant *p*-values.

Abbreviations: BMI, body mass index; BSF, bristol stool form; CTT, colonic transit time; GCSI, Gastroparesis Cardinal Symptom Index; GES, gastric emptying scintigraphy; GET, gastric emptying time by wireless motility capsule.

focus of the present analyses. In subjects with delayed compared to non-delayed GET, there were no differences in age, gender, diabetes status, baseline GCSI severity, functional dyspepsia, or constipation status by Rome III, BSF forms showing constipation at baseline, baseline CTT, and baseline constipation medication use within the different CTT subgroups (Table 1). Those with combined delayed CTT and delayed GET had a higher proportion of delayed gastric emptying by GES (*p* = 0.02) compared to those with delayed CTT with non-delayed GET. Among patients with delayed CTT, BMI was lower in those with delayed GET compared to non-delayed GET (*p* = 0.008). This was not seen in patients with normal CTT. There were no differences in specific constipation medications recommended by site investigators between those with delayed versus normal CTT (Table 2). Changes in constipation medications compared to baseline for individual subjects are shown in Table 3.

3.2 | Symptom Response

Patients enrolled in this study reported GCSI scores of 2.78 ± 0.95 at baseline, reflecting moderate severity of gastroparesis, and

constipation scores of 3.45 ± 1.46 , indicating moderate to severe constipation. Thirty-six of the 56 patients (64.3%) who were given either a new constipation medication or whose constipation therapy was changed exhibited delayed CTT > 59 h. Initiating new constipation therapy or changing constipation medication treatment reduced GCSI scores to 2.28 ± 1.15 (*p* = 0.002) and decreased constipation scores to 2.75 ± 1.10 (*p* = 0.004) in the whole group. In adjusted analyses, neither age, sex, BMI, neuromodulator use, prokinetic use, nor functional dyspepsia status contributed to significant GCSI and constipation score differences.

3.2.1 | Impact of Constipation Medication on UGI and Constipation Symptoms in Subjects With Delayed Colonic Transit

Thirty-six subjects with delayed CTT received a change in constipation medication from baseline, including 17 subjects (47.2%) who were taking a different constipation medication at baseline prior to enrollment. Overall GCSI scores improved significantly among all patients with delayed CTT at 6 months from baseline with a mean difference of -0.58 (*p* = 0.007) (Figure 1). Fifteen of

TABLE 2 | Constipation medications utilized to tailor management changes.

	Delayed CTT <i>n</i> = 36	Normal CTT <i>n</i> = 20	<i>p</i>
Linaclotide	14 (38.9%)	7 (35%)	0.77
Lubiprostone	5 (13.9%)	1 (5%)	0.30
Misoprostol	—	1 (5%)	n/a
Multiple ^a	4 (11.1%)	—	n/a
PEG-3350	7 (19.4%)	8 (40%)	0.10
Pyridostigmine	4 (11.1%)	2 (10%)	0.90
Undefined	2 (5.6%)	1 (5%)	0.93

Abbreviation: CTT, colonic transit time.

^aOne subject received linaclotide and magnesium citrate; the second subject received linaclotide and bisacodyl; the third subject received pyridostigmine and PEG-3350; the fourth subject received pyridostigmine and lubiprostone.

these 36 subjects (41.7%) who had concurrent GET delay in addition to their CTT delay demonstrated significant overall GCSI reductions over 6 months with a mean improvement of -0.96 ($p=0.004$) (Figure 2A). In contrast, overall GCSI scores failed to decrease significantly with a mean change of -0.30 ($p=0.29$) in the 21 patients with delayed CTT but non-delayed GET.

GCSI individual symptom subscore changes for nausea/vomiting ($p=0.02$) and early satiety/fullness ($p=0.002$) significantly decreased in those with combined delayed CTT and delayed GET, and there were trends to significant improvements in bloating/distention subscores ($p=0.06$) (Figure 3A–C). In contrast, upper abdominal pain showed no significant reductions ($p=0.15$) (Figure 3D). In the subgroup with delayed CTT but non-delayed GET, only the early satiety/fullness subscore showed a trend to improvement over 6 months ($p=0.07$) (Figure 3A–D).

Constipation scores improved significantly among all patients with delayed CTT, with a mean change of -0.81 ($p=0.004$) (Figure 4). Among those with delayed CTT, mean changes in constipation scores of -0.48 were not significant in patients with delayed GET ($p=0.25$) while reductions were significant for those with non-delayed GET, with a mean change of -1.1 ($p=0.008$) (Figure 4A). Separate subgroup analyses showed no differences in the percentages of patients with reductions in constipation scores in patients with delayed versus non-delayed GET (53.3% vs. 61.9%, $p=0.74$). Furthermore, the percentages of patients with improved constipation were similar in those with delayed versus non-delayed GET who were not taking constipation medications at baseline (66.7% vs. 70%, $p=1.00$).

3.2.2 | Impact of Constipation Medication on UGI and Constipation Symptoms in Subjects With Normal Colonic Transit

Twenty subjects with normal CTT received a change in constipation medication from baseline, including 7 subjects (35%) who were taking a different constipation medication prior to enrollment. Overall GCSI scores did not improve significantly among all patients with normal CTT, showing a mean change of -0.36

($p=0.14$) (Figure 1). Only 4 of these 20 subjects with normal CTT (20.0%) had delayed GET, while 16 (80.0%) exhibited non-delayed GET. Of patients with normal CTT, GCSI changes were not significant in those with either delayed GET (mean change $+0.19$, $p=0.76$) or non-delayed GET (mean change -0.50 , $p=0.07$) (Figure 2).

Among individual gastroparesis symptoms, only early satiety/fullness subscores showed significantly improved changes in patients with normal CTT and non-delayed GET ($p=0.02$) (Figure 3). In contrast, changes in early satiety/fullness scores in those with delayed GET and nausea/vomiting, bloating/distention, and upper abdominal pain subscores, irrespective of GET status, showed no improvement in those with normal CTT (all $p>0.05$).

Constipation scores did not improve from baseline to 6 months among all patients with normal CTT, with a mean change of -0.50 ($p=0.29$) (Figure 4). When stratified by GET subgroups, constipation scores did not significantly improve for patients with delayed GET (mean change 0.00 , $p=1.00$) or non-delayed GET (mean change -0.63 , $p=0.25$) (Figure 4).

4 | Discussion

We studied patients with symptoms of gastroparesis who underwent concurrent WMC and GES tests and were managed based on both results. Only subjects who received a new or a change in constipation therapy and completed symptom surveys quantifying changes over 6 months were included. We noted improved gastroparesis symptoms on constipation therapy only in those with delayed colonic transit. Further, overall GCSI and some individual gastroparesis symptoms selectively decreased in those with slow colonic transit with concurrent gastric emptying delays.

Constipation is common in those with gastroparesis symptoms, being moderate to very severe in 58% of patients from the NIH Gastroparesis Consortium [7]. Multiple studies note rates of slow colonic transit ranging from 18% to 42% alone or associated with gastric or small bowel delays in suspected gastroparesis, highlighting the prevalence of global dysmotility in this population [9, 10, 12, 13]. This opens the possibility that therapies targeting extra-gastric abnormalities could be prescribed. In one pediatric functional dyspepsia cohort with associated constipation, osmotic laxatives improved dyspepsia and accelerated gastric emptying [21]. As in that study, initiating a new constipation medication or changing therapy in our patients led to improved GCSI scores indicating that upper GI symptoms are ameliorated by treatments that act on the colon. GCSI benefits with constipation therapy were noted only for patients with delayed colonic transit, while those with normal colonic transit did not report GCSI decreases.

Of interest was our finding that GCSI decreases in those with slow colonic transit were within the group with delayed gastric emptying, while those with slow colonic transit but non-delayed emptying and those with normal colon transit did not improve. One can speculate that therapies other than those given for constipation may have led to the improvements in patients with combined gastric and colonic delays. We previously noted that

TABLE 3 | Individual constipation medication changes per subject.

Subject	Constipation status		Constipation medication	
	Rome III constipation	Delayed CTT	At baseline	Changed to
190003	No	Yes	PEG-3350	Linaclootide
190005	No	Yes	None	PEG-3350
190008	No	No	Misoprostol	Misoprostol dose increased
190013		No	None	Pyridostigmine
190014	No	No	Lubiprostone	PEG-3350
260001	Yes	Yes	None	Lubiprostone
260003	No	Yes	None	Pyridostigmine
260008	Yes	Yes	None	Pyridostigmine
260013	Yes	Yes	Bisacodyl, PEG-3350	Lubiprostone
260015	Yes	Yes	Bisacodyl	Pyridostigmine
260017	Yes	No	None	Pyridostigmine
260021	Yes	Yes	Magnesium oxide	Lubiprostone
284001	No	Yes	None	PEG-3350
284006	No	Yes	PEG-3350	Linaclootide
284007	No	No	PEG-3350 as needed	PEG-3350 daily
284009	No	Yes	None	Lubiprostone
284012	No	No	Docusate	PEG-3350
284013	No	Yes	PEG-3350	Linaclootide
284014	No	No	Magnesium oxide, PEG-3350	Linaclootide
286002	Yes	No	None	PEG-3350
286005	Yes	Yes	None	Linaclootide
286008	No	No	None	PEG-3350
286009	No	No	None	PEG-3350
286010	No	Yes	None	PEG-3350
286014	Yes	Yes	None	PEG-3350
286016	No	No	Lubiprostone	Linaclootide
286018	Yes	Yes	PEG-3350	Linaclootide
286025	No	No	None	PEG-3350
286027	No	Yes	Pyridostigmine	Pyridostigmine dose increased
286028	No	Yes	Bisacodyl	Linaclootide
286030	No	Yes	Linaclootide	Linaclootide dose increased
287003	Yes	No	None	PEG-3350
287010	Yes	Yes	None	PEG-3350
287011	Yes	Yes	None	PEG-3350
289001	Yes	Yes	None	Linaclootide, bisacodyl
289002	No	No	None	Linaclootide
289005	No	No	Docusate	Linaclootide

(Continues)

TABLE 3 | (Continued)

Subject	Constipation status		Constipation medication	
	Rome III constipation	Delayed CTT	At baseline	Changed to
289007	No	No	None	Linacotide
289008	No	Yes	Lubiprostone	Linacotide
289010	No	Yes	PEG-3350	Linacotide
289011	Yes	Yes	None	Linacotide, Magnesium citrate
289014	No	Yes	None	Undefined
289015	Yes	Yes	Linacotide	Linacotide dose increased
289016	No	Yes	None	Linacotide
289020	No	Yes	None	Linacotide
290001	Yes	Yes	PEG-3350	Undefined
290002	Yes	Yes	Colace	PEG-3350
290003	Yes	Yes	None	PEG-3350
290007	Yes	No	None	Undefined
291010	No	No	None	Lubiprostone
291014	No	Yes	Bisacodyl	Lubiprostone
291015	No	Yes	None	Linacotide
306001	Yes	No	None	Linacotide
306002	Yes	No	None	Linacotide
306007	No	Yes	Linacotide	Linacotide dose increased
306008	Yes	Yes	None	Lubiprostone

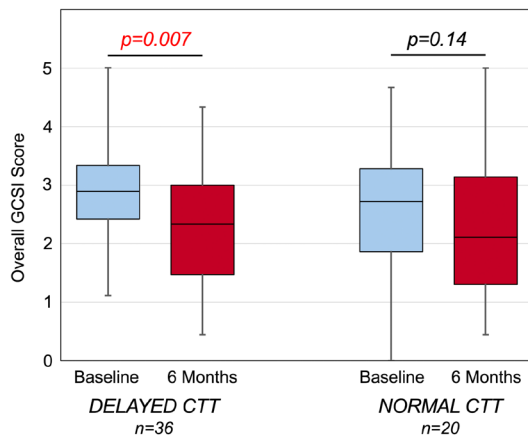


FIGURE 1 | Overall GCSI scores significantly decreased from baseline to 6 months in patients with delayed ($p = 0.007$) but not normal CTT ($p = 0.14$). Data are presented as median (IQR) with maximum and minimum values.

prokinetic agents are preferentially given for gastric emptying delays, but we did not see consistent efficacy of prokinetics in reducing gastroparesis symptoms [16, 22]. Conversely, neuromodulators plus prokinetics in combination or neuromodulators alone elicited benefits. However, in our generalized linear model, adjusting for prokinetics did not impact responses to constipation therapy. Some constipation therapies also exert prokinetic

action in the stomach. We classified pyridostigmine as constipation therapy, but cholinesterase inhibitors can accelerate gastric emptying [23]. Only 2 of 6 patients on pyridostigmine exhibited combined gastric and colon delays, and no patient received prucalopride in this study.

Improvements in nausea/vomiting paralleled overall GCSI reductions, decreasing only with both gastric and colonic delays. Early satiety/fullness improved in all patients except for those with delayed gastric emptying and normal colon transit, likely due to the small sample in this group. The broad-based response of fullness to constipation therapy irrespective of gastric and colon transit suggests a lesser role for transit in the pathogenesis of this symptom. Bloating/distention trended to improvement with combined gastric and colonic delay, while pain did not improve in any group. Our findings mimicked other studies that quantified individual symptom responses to therapies in gastroparesis and related disorders. In one study, bloating/distention severity predicted the lack of response to antiemetics, prokinetics, and neuromodulators [24]. In another report, gastric electrical stimulation for gastroparesis relieved nausea/vomiting but not pain [25].

Constipation was moderate to severe in our patients who were given new constipation medications or a change in constipation therapies. Nearly two-thirds of these patients exhibited delayed colonic transit. A subset of subjects with normal colonic transit

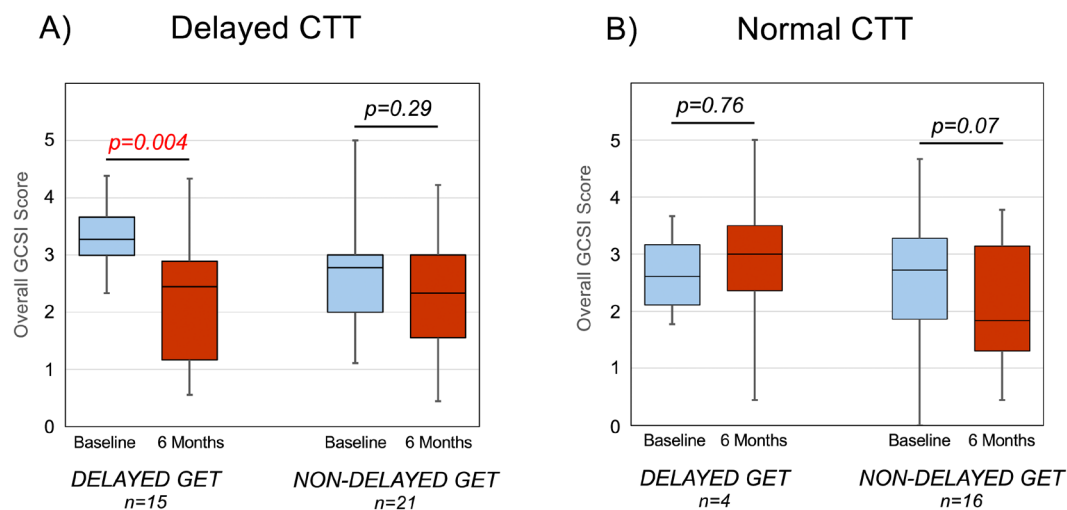


FIGURE 2 | Overall GCSI scores at baseline and 6 months are shown for patients in relation to transit delays in the colon and stomach. Of patients with delayed CTT (A), only those with associated delayed GET showed significant GCSI reductions scores at 6 months ($p=0.004$) while those with non-delayed GET did not improve ($p=0.29$). Of patients with normal CTT (B), neither those with delayed GET ($p=0.76$) nor non-delayed GET ($p=0.07$) showed reduced GCSI scores. Data are presented as median (IQR) with maximum and minimum values.

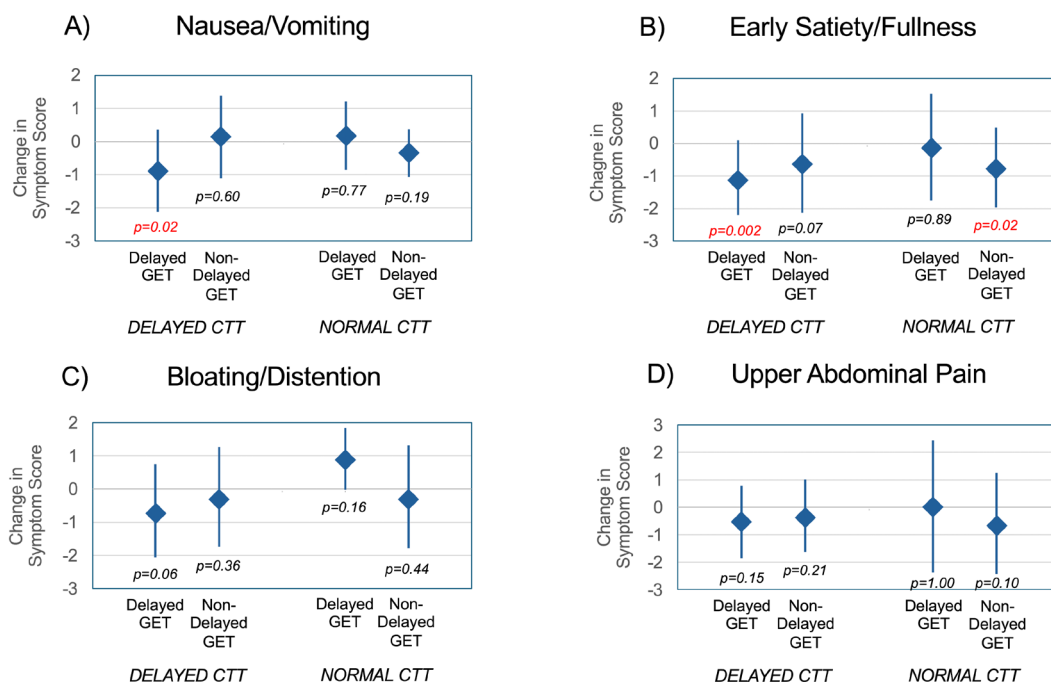


FIGURE 3 | Changes in individual GCSI symptom subscores and upper abdominal pain scores from the modified PAGI-SYM survey from 6 months minus baseline values are shown. Nausea/vomiting subscores decreased over 6 months only in patients with combined CTT and GET delays ($p=0.02$) but did not improve in other patient groups (A). Early satiety/fullness subscores decreased significantly in those with combined CTT and GET delays ($p=0.002$) as well as patients with normal CTT and non-delayed GET ($p=0.02$) and showed trends to reductions in those with delayed CTT but non-delayed GET ($p=0.07$) (B). Bloating/distention subscores showed trends to reductions only with combined CTT and GET delays ($p=0.06$) (C). Upper abdominal pain scores did not significantly decrease in any group (D). Data are presented as mean change in score $\pm$ SD.

that neither met Rome III constipation criteria nor reported BSF types 1 or 2 received constipation therapy changes. Of note, these patients often reported constipation scores of moderate or greater severity on the modified PAGI-SYM survey, perhaps reflecting imperfect sensitivity of the Rome and BSF measures in constipation similar to other studies reporting 55%–70%

sensitivity of the Rome criteria in IBS or functional dyspepsia [26–28]. Constipation scores decreased in those with delayed but not normal colon transit. This raises the possibilities that delayed colon transit predicts relief of constipation using therapies targeting the colon in patients with suspected gastroparesis, while treatments that address other pathogenic factors may be

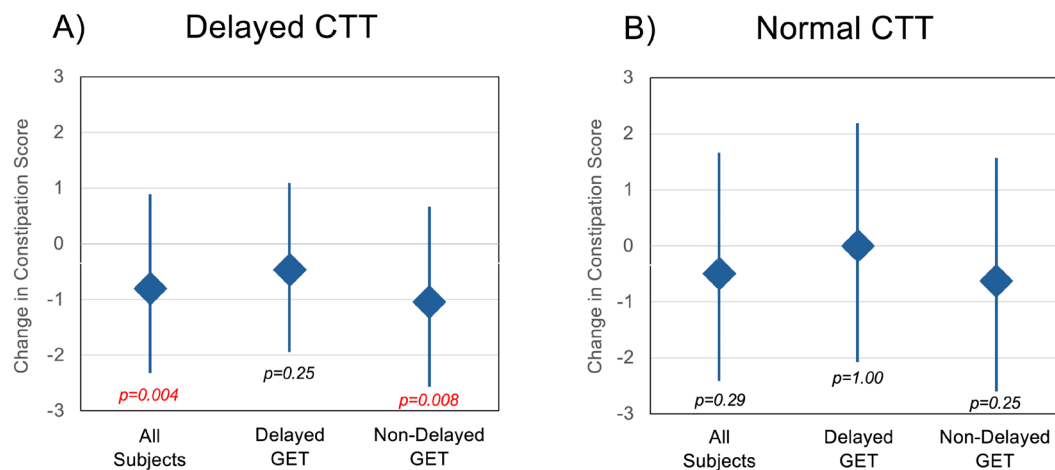


FIGURE 4 | Changes in constipation scores from the modified Pagi-SYM survey from 6 months minus baseline values are shown. Among patients with delayed CTT, constipation scores decreased significantly ($p=0.004$) (A). In the group with delayed CTT, those with non-delayed GET also showed constipation score reductions ($p=0.008$) while individuals with delayed GET did not ($p=0.25$). Overall, patients with normal CTT did not exhibit significant constipation score decreases ($p=0.29$) (B). Of those with normal CTT, neither patients with delayed ($p=1.00$) or non-delayed ($p=0.25$) showed improved constipation scores. Data are presented as mean change in score $\pm$ SD.

required for those with normal colonic transit. Numerical reductions were seen with both delayed and non-delayed gastric emptying, but improvements were statistically significant only in those without stomach emptying delays. However, on subgroup analyses, comparable percentages of those with delayed versus non-delayed gastric emptying showed reduced constipation severity, suggesting the apparent differences in response may be influenced by small sample sizes. As all patients received a change in constipation therapy, it was surprising that the observed constipation improvements were modest. We recognize that the Pagi-SYM was designed to assess the severity of gastroesophageal reflux, dyspepsia, and gastroparesis; thus, this was an appropriate instrument for our cohort, which was recruited to evaluate gastroparesis symptoms [20]. Other surveys, like the Patient Assessment of Constipation Symptoms, are better constructed to characterize constipation severity using 12 questions rather than a single item, as in the Pagi-SYM [29]. Thus, it is possible that our survey was inadequately sensitive to measure changes in constipation in response to therapeutic interventions. Bristol Stool Form scores also provide insight into constipation severity, but we did not quantify Bristol score changes at 6 months [18]. Despite potential drawbacks of our constipation survey, it is intriguing that constipation reductions were not greater in the subgroup of patients with combined gastric and colon delays who exhibited prominent gastroparesis symptom improvements, suggesting that GCSI and constipation score decreases may be independent of each other.

Physiologic actions of constipation therapy to benefit gastroparesis symptoms in those with combined gastric and colon transit delays are poorly understood. Mechanisms other than emptying delays influence gastroparesis symptoms, including blunted accommodation, hypersensitivity, mucosal inflammation, and gut dysbiosis, but most constipation medications are not known to exert actions on these pathways [30]. In theory, actions of constipation therapies to accelerate colonic transit and promote fecal expulsion might reduce gastroparesis symptoms by relieving functional outlet obstruction. Reflexes linking gastric

activity with colon function are described. In healthy subjects, volitional withholding of defecation can delay gastric emptying [31]. Likewise, rectal distension in normal controls slows gastric emptying and prolongs duodeno-cecal transit [32, 33]. These studies highlight the colo-gastric brake—the reflex inhibition of upper GI motility by colonic distention [21]. Such mechanisms may reduce upper GI symptoms in patients with delayed gastric and colon transit because they stimulate fecal expulsion and inactivate colo-gastric brake reflexes. In contrast, colo-gastric brake reflexes may not be activated in those with normal colon transit, and additional stimulation of colon propulsion may not further accelerate gastric emptying.

Our analyses suggest that evaluating whole gut transit may offer management insight beyond traditional approaches that include unguided empiric treatment of symptoms or therapy directed solely to the rate of gastric emptying. While study investigators had both GES and WMC results available to review before enacting management changes, the decision for constipation therapies was mostly influenced by the whole gut WMC transit results [16]. Agents targeting constipation were beneficial only in patients with combined delays in gastric emptying and colon transit in our study, but other therapies might be more appropriate for those with non-delayed gastric emptying and/or normal colonic transit. In another publication on this cohort, we described selective benefits of neuromodulators only in patients with non-delayed gastric emptying—a group distinct from those showing improvements on constipation therapies in this study [22]. Such medications might also be employed for patients with normal colonic transit. Benefits of constipation medications coupled with therapies with other mechanisms of action might be additive, but this was not analyzed in our study because of power limitations. Measuring colonic transit provided insight into responses to constipation therapy, but we note the WMC technology in this study is no longer commercially available. The WMC demonstrates comparable sensitivity and specificity to radiopaque markers testing to predict constipation symptoms (AUC of the ROC Curve = 0.73, sensitivity 0.46, specificity 0.95) with

significant correlation between the two testing modalities in evaluating colonic transit ($r=0.71$, $p<0.001$) [34, 35]. Therefore, radiopaque markers can also provide a means of evaluation accessible to most clinical practices. Alternatively, colonic transit can be discerned using whole gut scintigraphy in specialty centers, as well as newer capsule methods of transit testing that are anticipated and currently undergoing validation [36].

Our study had limitations. This was an observational cohort study and not a randomized nor case-controlled trial, limiting our ability to control for factors like post-testing therapies other than constipation medications. Although data were collected prospectively, this study was a secondary analysis that was not optimally powered to assess the impact of constipation therapies in all subgroups. Interpreting results in the normal colonic transit group with delayed gastric emptying was particularly limited by its small sample size. Likewise, we could not perform many between-group comparisons or assess impacts of individual constipation medications on outcomes. We did not confirm adherence to constipation therapy at 6 months, but it is likely that differences in responses between groups would not have been seen if patients were noncompliant. These analyses provided incomplete mechanistic explanations for our findings. WMC tests were not repeated at 6 months of therapy, which would have correlated reduced symptoms to improvements in gastric and colonic transit. We did not consider or treat other mechanisms of constipation, including dyssynergic defecation [37, 38]. Further, none of the constipation management changes included rectal therapies such as bisacodyl suppositories, enemas, or water irrigation. Subgroup characterizations observed differences in BMI, being lower in those with combined gastric and colon delays. BMI is reported as a risk for constipation and predicts response to laxatives, but lower BMI did not influence any response to constipation therapy in our patients [39, 40]. Despite these limitations, we believe this pragmatic prospective cohort approach permitted the generation of believable data to demonstrate that targeting therapies for colonic dysmotility can improve upper GI symptoms in selected settings.

In summary, our study builds upon prior reports that characterized constipation in patients with symptoms of gastroparesis. We further found that tailored therapy of constipation in patients with delayed colon transit improves global gastroparesis symptoms, particularly with concomitant gastric emptying delays. Individual symptoms that benefited most from constipation therapy included nausea/vomiting and early satiety/fullness, with lesser reductions in bloating/distention and no improvement in abdominal pain. Our findings suggest potential clinical utility in evaluating whole gut transit in individuals with symptoms suspicious for gastroparesis to identify patient subsets with colonic dysmotility who may be responsive to therapies that target the colon.

Author Contributions

B.C.S., B.M., W.L.H., and S.S.C.R. were involved in study design, interpretation of the data, and writing of the report. B.C.S. analyzed the data. B.M., A.A.L., R.W.M., I.S., L.A.N., M.I.S., J.M.W., H.P.P., B.K., W.L.H., and S.S.C.R. collected the data. All authors provided revisions and approval of the submission.

Ethics Statement

The protocol was approved by the Institutional Review Board at each center.

Consent

Informed consent was obtained from each subject before enrollment.

Conflicts of Interest

B.M., A.A.L., R.W.M., I.S., L.A.N., M.I.S., J.M.W., H.P.P., B.K., W.L.H., and S.S.C.R. were funded site Principal Investigators for the parent investigation of this study. B.C.S. has no conflicts to disclose.

Data Availability Statement

Deidentified individual participant data that underlie the reported results will be made available from the corresponding author upon reasonable request.

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